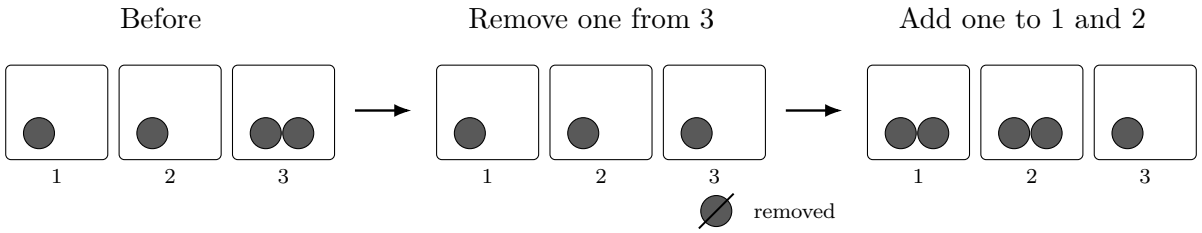
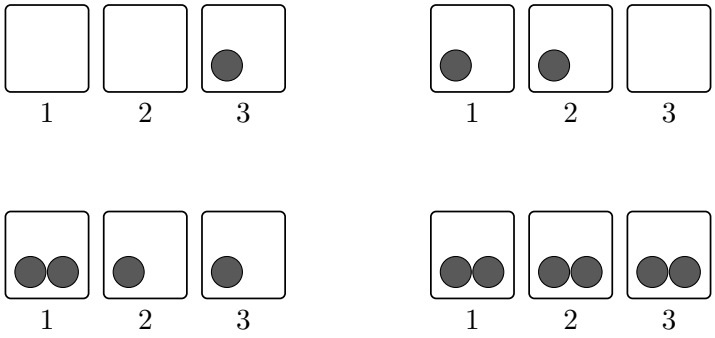
Take turns. Remove one counter from a nonempty bucket. You may then add counters only to buckets strictly to its left. If you cannot move, you lose.

For Problems 1–3, add at most one counter to each bucket to the left of the one you took from.



Problem 1: Play each start with a partner. Which starts let the first player win whatever the second player does?

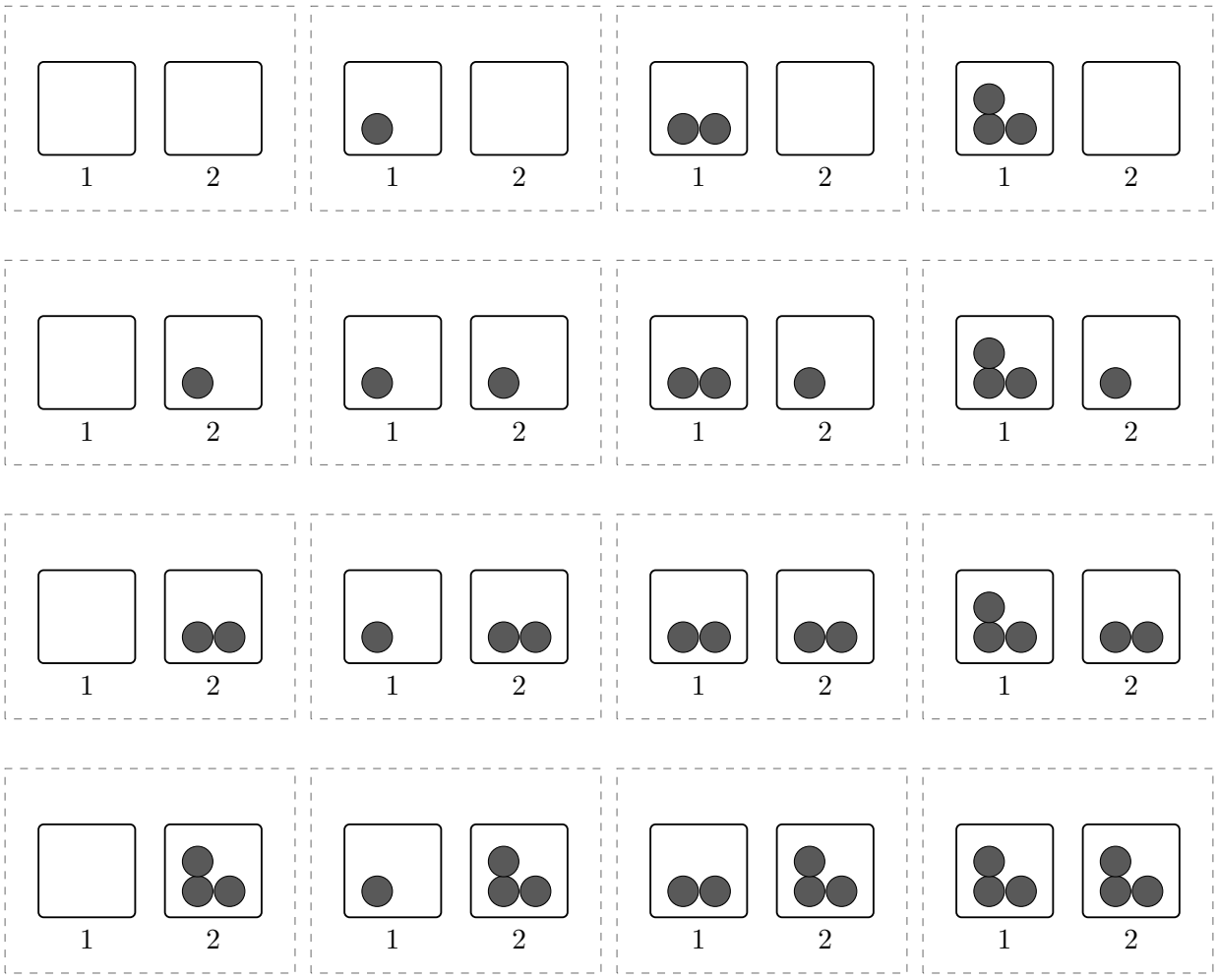


Bucket 1 | Bucket 2 | Bucket 3

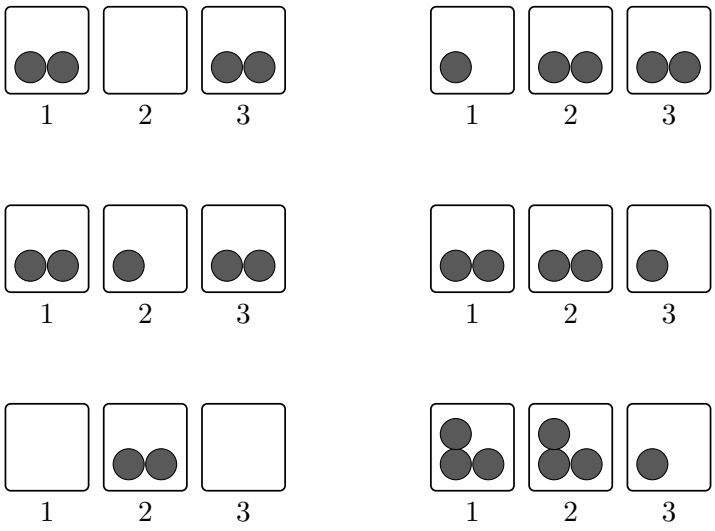
Supply

Removed

Problem 2: Use two buckets. Sort these starts into two groups: the first player can force a win, or the second player can force a win.

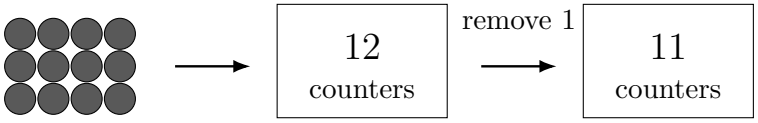


Problem 3: Play from these three-bucket starts. Can you find a rule for choosing a winning move whenever there is one?

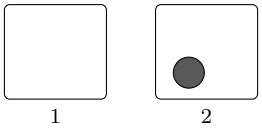


For Problems 4–5, a refill may be any finite number of counters. It still goes only to buckets strictly to the left of the bucket you took from.

When playing, you may write a number to stand for that many counters.



Problem 4: Start with an empty first bucket and one counter in the second. Work together to make a game last at least 10 moves, then 30 moves, then 100 moves. Can you beat any finite number someone names?



Problem 5: Start from each picture. Can you agree on moves that keep the game going forever? If you think every game ends, however big the refills, explain why.

